PHYSCI 101 LAB: PHYSICAL SCIENCE LABORATORY INVESTIGATIONS

Teresa Ciardi
College of the Canyons

LibreTexts™

Physical Science 101 Lab

The LibreTexts libraries are Powered by NICE CXone Expert and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

This text was compiled on 08/11/2026

TABLE OF CONTENTS

Licensing

A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing****.*

CHAPTER OVERVIEW

1: The Scientific Method

The scientific method has been described in numerous textbooks as a set of steps to be followed. Steps may be memorized without understanding the true nature of the steps, the true nature of doing science. The way in which a scientist approaches answering an unknown is through informed experimentation. Scientists approach the unknown as a question to be answered and devise a method, based on prior knowledge and experience, to answer the question. The method relies on precise measurements which allows for the results to be verified. Because scientists are dealing with an unknown, something for which there is not an answer, they must often begin with a hypothesis. A hypothesis is what the informed scientists thinks the answer to the question will be; it is essentially an educated guess.

- Teresa Ciardi

1.1: Interpreting Errors

Systematic Error

- Measurements deviate from the correct value by the same amount each time
- Errors have a pattern

Example: Time on a watch is always 5 minutes ahead of the true time

Random Error

- Errors have no pattern

Example: Everyone's watch shows a different time

Uncertainty in Measurement

- Number ± error
- It is the amount a number may be off from the correct value

Example: Age ±365 days

Ratio Comparison

- Shows how two numbers compare
- Reduced fraction shows comparison

Example:

$$\frac{65\text{mph}}{30\text{mph}} = \frac{2.2}{1} \text{ (Means one speed is 2.2 times faster than the other)}$$

Percent Error

- Percentage measurements are off from standard/correct value
- Absolute value

$$\text{Percent Error} = \frac{\text{Experimental/Measured Value} - \text{Accepted/Theoretical Value}}{\text{Accepted/Theoretical Value}} \times 100\%$$

CHAPTER OVERVIEW

2: Investigation 1 - Making a Hypothesis

Learning Objectives

- Experience making a hypothesis, taking good measurements, and developing a procedure to solve a problem.
